EXPERIMENT 9

Aim: Create an ERC 20 Token in Remix IDE

THEORY

1. What is a Token?

A token is a digital asset created and managed using a blockchain's smart contract functionality. Tokens can represent currencies, assets or utilities.

Common token standards include:

- ERC-20: Fungible tokens.
- ERC-721: Non-fungible tokens (NFTs).
- ERC-1155: Supports both fungible and non-fungible tokens.

2. What is ERC-20?

ERC-20 (Ethereum Request for Comment 20) is a standard for creating fungible tokens on the Ethereum blockchain. Each token is interchangeable with another token of the same type.

It defines common functions and events that allow wallets, exchanges and DApps to interact with tokens.

3. Remix IDE

Remix IDE is a browser-based development environment used to write, compile, deploy and test smart contracts written in Solidity.

It provides tools for compiling contracts, connecting wallets and deploying smart contracts on blockchain networks.

4. MetaMask

MetaMask is a cryptocurrency wallet used to manage Ethereum accounts and interact with decentralized applications. It connects Remix IDE to the blockchain and is used to approve transactions and pay gas fees.

5. Sepolia Testnet

Sepolia is an Ethereum test network used for developing and testing smart contracts. It uses test ETH instead of real ETH to perform transactions and pay gas fees.

6. ERC-20 Token Functions

Important ERC-20 functions include:

Function	Purpose
`name()`	Returns token name
`symbol()`	Returns token symbol
`decimals()`	Returns decimal precision
`totalSupply()`	Returns total token supply
`balanceOf()`	Checks account balance
`transfer()`	Transfers tokens
`approve()`	Approves token spending
`allowance()`	Checks approved spending limit
`transferFrom()`	Transfers approved tokens

7. ERC-20 Events

Events record important activities performed by a smart contract.

- Transfer: Emitted whenever tokens are transferred.
- Approval: Emitted whenever an owner approves another account to spend tokens.

8. ERC-20 Attributes

- Name: Full name of the token.
- Symbol: Short representation of the token.
- Decimals: Number of decimal places supported, commonly 18.
- Total Supply: Total number of tokens created.
- Balance: Number of tokens held by an account.

9. ERC-20 vs ERC-721

ERC-20	ERC-721
Fungible tokens	Non-fungible tokens
Tokens are interchangeable	Each token is unique
Used for currencies and utility tokens	Used for NFTs
No unique ID for each unit	Each token has a unique ID

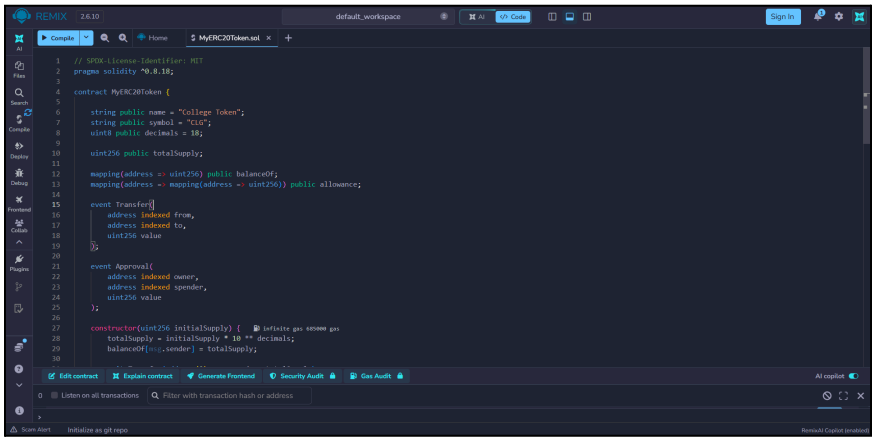
10. ERC-20 Extensions

Mintable: Allows creation of new tokens, increasing the total supply.

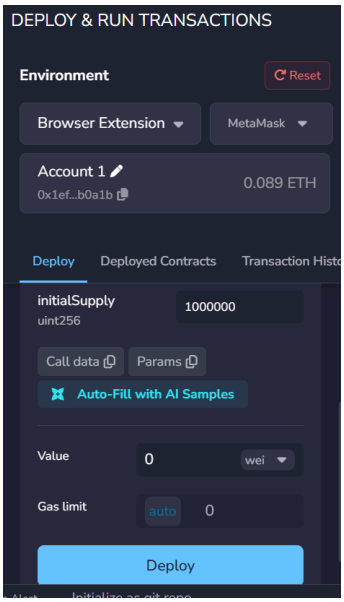
Burnable: Allows permanent destruction of tokens, decreasing the total supply.

Pausable: Allows authorized accounts to temporarily stop token transfers during emergencies.

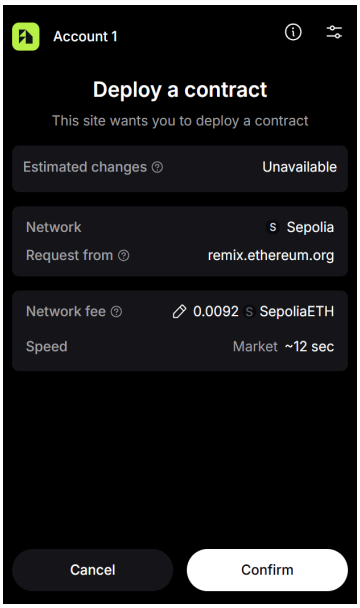
Output & Code:



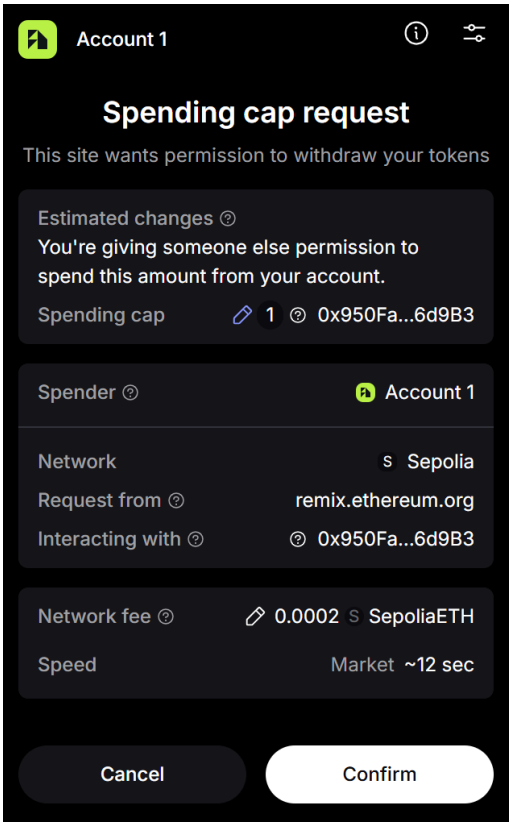
Smart Contract for token creation and transfer



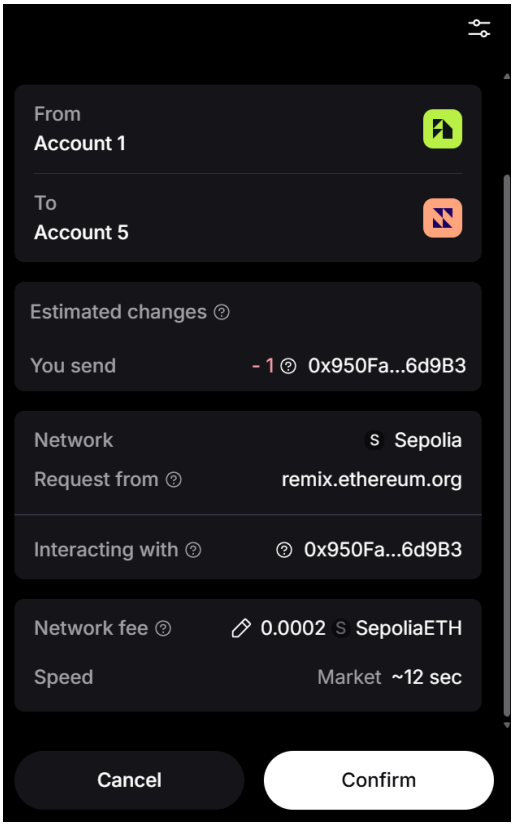
Initial Supply of tokens



MetaMask Connection



Approval for account 2 to transfer tokens



Transferring tokens from Acc 1 to Acc 5

Conclusion

Thus, the ERC-20 token was successfully created using Solidity in Remix IDE and deployed on the Sepolia Testnet using MetaMask. The token was assigned an initial supply of 1,000,000 College Tokens (CLG) with 18 decimals. The experiment demonstrated token creation, deployment, approval and transfer operations, providing practical understanding of ERC-20 standards and smart contract interactions.